

Option B: Coagulant Dosing at a Water Treatment Plant

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1. Problem formulation

Decision and audience. Every hour, a shift operator chooses an alum dose (mg/L) from four observed conditions. Underdosing reduces the safety margin on treated-water turbidity; overdosing raises chemical use and sludge. The system therefore has to provide a numeric setpoint while remaining explainable rule-by-rule.

Variable	Synthetic range	Role in model
Raw-water turbidity	2-150 NTU	Primary solids/particle-load signal
pH	5.5-8.5	Dose increases as pH moves away from the synthetic optimum near 7.0
Inflow rate	20-120 MLD	Higher throughput increases required dose
12-hour rainfall	0-120 mm	Catchment disturbance / event intensity proxy
Output	5-90 mg/L	Recommended alum dose for the next hour

2. Synthetic data and known ground truth

We generated 420 samples (seed 20260909) because the brief explicitly permits a documented ground-truth model plus noise. Input distributions combine clipped log-normal turbidity, near-neutral pH, variable inflow and zero-inflated rainfall with occasional storm events. Gaussian output noise has SD 2.2 mg/L.

With normalized turbidity t , flow f and rainfall r in $[0,1]$, and pH penalty $q = \min(|pH-7.0|/1.5, 1.2)$, the noiseless synthetic target is:

$$\text{dose} = \text{clip}[8 + 42\sqrt{t} + 8f + 12r + 14q + 10tr + 4fr, 6, 86] \text{ mg/L}$$

This equation is a controlled coursework model, not a calibration claimed for a real Malaysian plant. Its purpose is to create a nonlinear but auditable relationship against which optimization error can be measured.

3. Search-space arithmetic

Chromosome length = (4 inputs $\times$ 3 membership-function knots) + 36 zero-order TSK consequents + 27 optional rule gates = 12 + 36 + 27 = 75 genes. The 27 binary gates alone create $2^{27} = 134,217,728$ possible rule subsets before continuous membership/consequent tuning.

4. Fuzzy-system design

Choice: zero-order Takagi-Sugeno-Kang (TSK). The output is consumed as an exact mg/L setpoint, so constant consequents avoid a second defuzzification approximation. Interpretability is preserved because each rule still has linguistic antecedents and prints a single numeric dose. This is preferable here to a black-box regressor because the 3 a.m. operator must be able to defend the recommendation.

Each input uses Low / Medium / High sets. A decoder sorts and spaces the three knots so semantic order cannot invert. Low and High are shoulder functions; Medium is triangular. The nine Turbidity $\times$ pH rules are mandatory and use wildcards for flow/rainfall, guaranteeing base coverage. Twenty-seven additional interaction rules can be switched off by evolution.

Encoding choice. The chromosome is mixed: real-valued membership knots and rule consequents, with binary gates for the optional rules. The operators respect that boundary — arithmetic blend and Gaussian nudges on the real segment, uniform selection and bit flips on the gate segment. The alternative we rejected was a fully binary encoding in which every real parameter is quantised into bits; we rejected it because flipping a high-order bit moves a dose by tens of mg/L, so neighbouring bit strings are not neighbouring controllers.

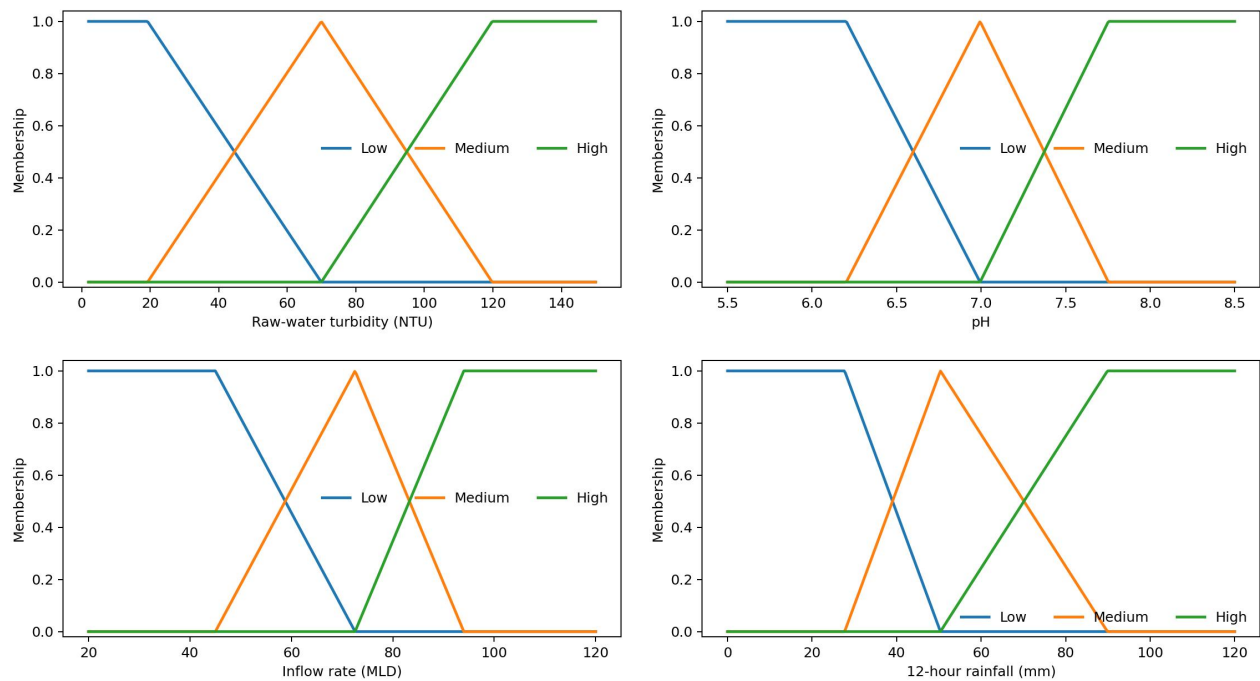


Figure 1. Membership functions of the shipped knee solution. Knots are evolved, but decoder guards preserve order and overlap.

Input	Coverage	Min $\Sigma\mu$	Max adjacent overlap	Normalized knots
turbidity_ntu	100.0%	1.000	0.500	0.116, 0.459, 0.796
ph	100.0%	1.000	0.498	0.234, 0.498, 0.751
inflow_mld	100.0%	1.000	0.499	0.250, 0.526, 0.740
rainfall_12h_mm	100.0%	1.000	0.498	0.231, 0.420, 0.749

Mechanical coverage/overlap check on a 401-point grid per input; no assertion is made without a computed check.

5. Inference and decoder guards

For sample x , rule firing strength is the product of the memberships of its non-wildcard antecedents. The TSK output is $y(x)=\sum w_i(x)c_i / \sum w_i(x)$. Mandatory base rules make the denominator non-zero throughout the input domain.

- Guard 1: each input knot triple is sorted and constrained to $[0.08, 0.92]$.
- Guard 2: adjacent knots must be at least 0.12 apart; excessive shifts are projected back into bounds.
- Guard 3: the first nine coverage rules can never be deleted; only 27 correction rules are gated.
- Guard 4: rule consequents are clipped to 8-86 mg/L.

6. Hybrid evolutionary tuning

NSGA-II was implemented from scratch: non-dominated sorting, crowding distance, binary tournament selection, arithmetic/uniform crossover, mutation and elitist environmental selection. No optimization library is used. The same chromosome and decoder are used by random search and the scalarised GA comparators.

Two objectives

- Objective 1 - Accuracy: RMSE in mg/L on the training split; lower is better.
- Objective 2 - Interpretability cost: active-rule count + $1.5 \times$ (normalized MF semantic drift + normalized consequent drift); lower is simpler and closer to the expert linguistic partition.

The active-rule term dominates the second objective. Semantic-drift penalties prevent an apparently compact rule base from becoming hard to name because its fuzzy sets have moved to unintuitive positions.

Sign convention. Both objectives are written as minimisations; nothing is maximised anywhere in the search, so no sign is flipped inside the dominance test and the crowding-distance normalisation is applied per objective.

Constraint policy. Coverage is a constraint, not a third objective. It is enforced first in the representation — the nine mandatory Turbidity x pH rules carry wildcards for flow and rainfall, so an input with zero total firing strength cannot occur — and only as a backstop by a penalty of 50 mg/L per unit of uncovered fraction. That penalty never triggered in any run reported here, so neither a repair operator nor rejection of infeasible candidates was needed.

Component	Remove it and this happens
Tournament selection	No preferential reproduction, so there is no pressure towards better regions; the run degrades into a random walk.
Crossover	Only parallel hill climbing remains; the one mechanism that combines two partial solutions is gone.
Mutation	Only the material present in the initial population can ever be recombined; once diversity is exhausted the run stalls above the achievable optimum.
Elitism (parent + child merge)	A rank-1 solution can be destroyed by its own children, and the front can finish worse than it was halfway through.
Crowding distance	The surviving front clusters into one corner: still non-dominated, but no longer a usable menu.

Table 2. Component removal: remove one line of the canonical loop and this is what breaks.

Elitism is structural: parents and children are merged before environmental selection, so a rank-1 solution cannot be lost. We report the knee of the final merged front and never read the best of the last generation as the result of the run.

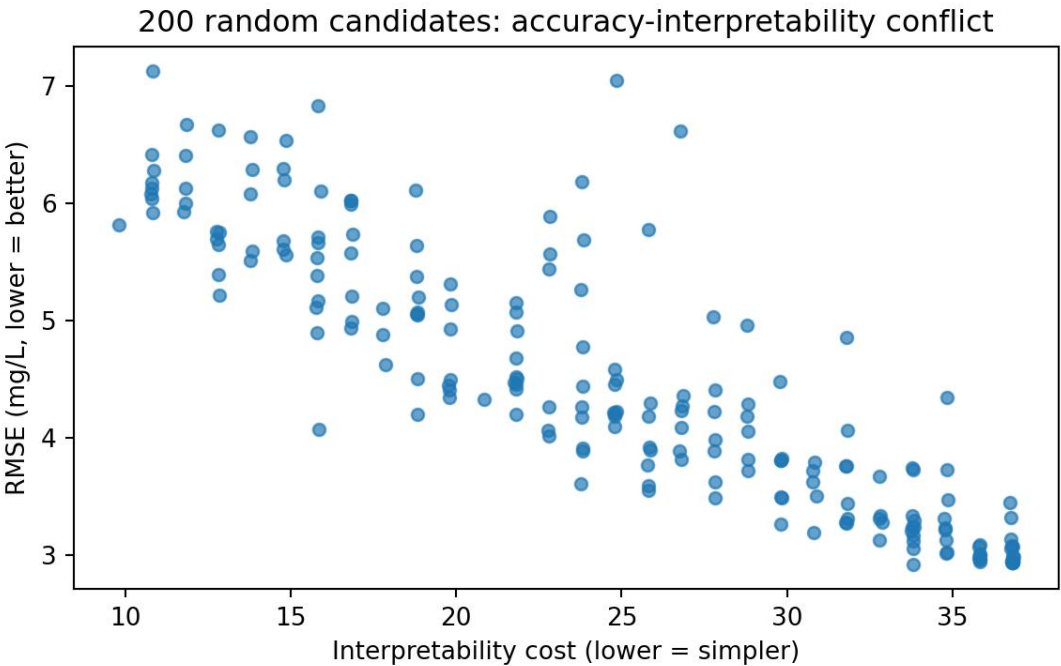


Figure 2. 200 randomized candidates across complexity levels. Spearman $\rho = -0.898$, $p = 2.00e-72$. Lower interpretability cost tends to require higher RMSE, establishing a real conflict.

Protocol and equal budget

Protocol item	Value
Dataset	420 synthetic observations; fixed generator seed 20260909
Evaluation	30 independent 70/30 train-test splits

Search population	24
Generations	12 after initial population
Budget	$24 \times (1 + 12) = 312$ candidate evaluations per search method
Random search	312 independently randomized candidates
Scalarised GA	RMSE + $0.10 \times$ interpretability cost
NSGA-II	Two objectives kept separate; knee selected after the front is produced

7. Benchmarking results

All values below are held-out test RMSE across 30 seeds. The untuned expert controller is evaluated without search; random search, scalarised GA and NSGA-II receive the same 312-evaluation budget. Polynomial ridge is a deterministic non-fuzzy baseline fitted on each training split.

Method	Test RMSE (mg/L), mean $\pm$ SD	Interpretability cost, mean $\pm$ SD
Untuned expert fuzzy	6.356 ± 0.334	36.00 ± 0.00
Random search	5.460 ± 0.684	16.82 ± 5.23
Scalarised GA	4.296 ± 0.380	17.07 ± 2.02
Polynomial ridge	2.155 ± 0.101	-
NSGA-II TSK (knee)	4.804 ± 0.471	14.43 ± 1.77

Table 1. 30 seeds; 70/30 splits; population 24×12 generations = 312 evaluations for random search, scalarised GA and NSGA-II; data seed 20260909; output noise SD 2.2 mg/L; scalarisation $\lambda = 0.10$; interpretability cost = active rules + $1.5 \times$ normalised semantic drift. Lower is better for both reported metrics.

Best over the 30 seeds. The lowest test RMSE seen in any single seed was 3.933 mg/L for the NSGA-II knee, 3.464 for the scalarised GA, 4.429 for random search, 5.807 for the untuned expert system and 1.984 for the polynomial ridge. One run of a stochastic method is one sample; the mean and SD above, not these best values, carry the claim.

Statistical comparison

Paired comparison over 30 seeds	Wilcoxon W	Two-sided p
NSGA-II vs Untuned expert fuzzy	0	$1.863e-09$
NSGA-II vs Random search	39	$1.397e-05$
NSGA-II vs Scalarised GA	44	$2.689e-05$
NSGA-II vs Polynomial ridge	0	$1.863e-09$

Interpretation. NSGA-II is significantly better than the untuned expert system and random search, but the scalarised GA attains lower RMSE because it collapses both goals into one preference before searching. NSGA-II deliberately preserves alternatives and returns a lower-complexity knee. The polynomial ridge baseline is substantially more accurate (≈ 2.16 mg/L), so the fuzzy hybrid should not be sold as the best pure predictor. That negative result is informative rather than a failure: the fuzzy system earns its place through a compact, inspectable decision policy. A real deployment decision would therefore depend on how much accuracy the operator is willing to exchange for a rule base that can be reviewed and defended.

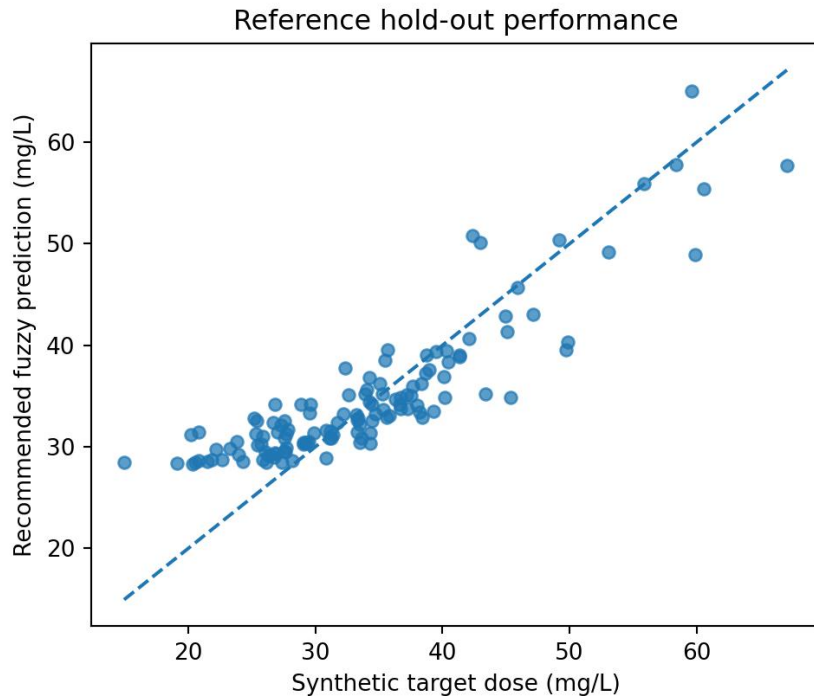


Figure 3. Reference hold-out predictions for the selected knee; test RMSE = 4.666 mg/L.

8. Pareto front and recommended operating point

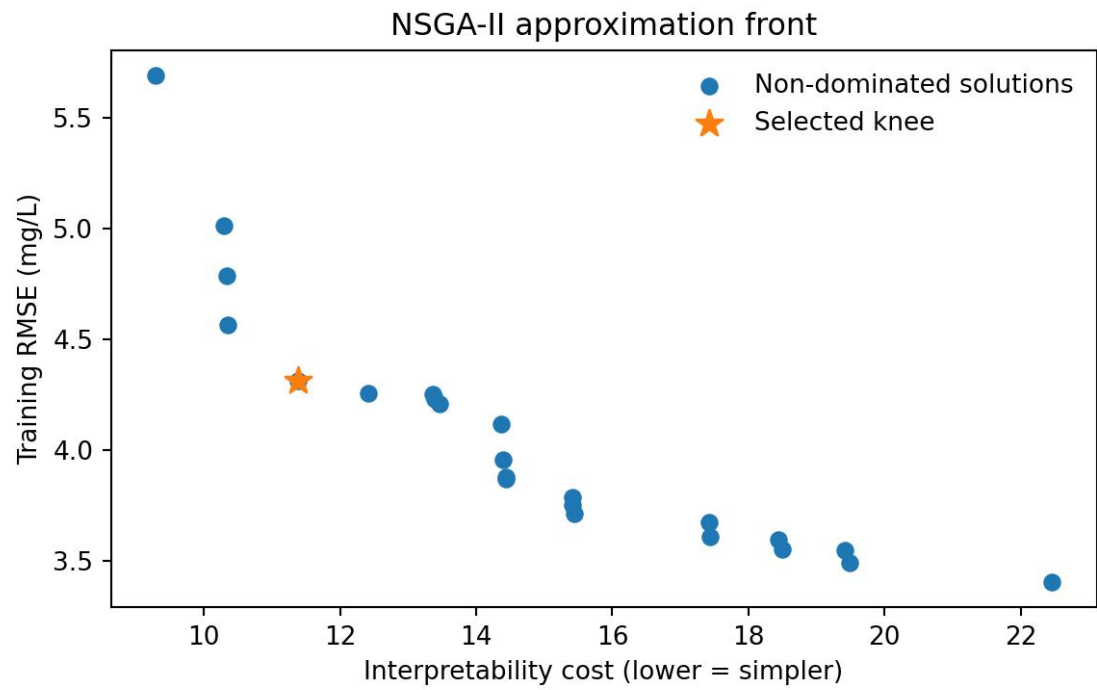


Figure 4. High-budget reference NSGA-II run (48 population, 50 generations). The star is the geometric knee of the non-dominated front. It is an approximation front: the non-dominated set found within one budget, not the true front.

Point	Train RMSE (mg/L)	Interpretability cost	Note
1	3.404	22.46	most accurate end
2	3.490	19.48	
3	3.548	19.42	
4	3.552	18.49	
5	3.596	18.44	
6	3.607	17.44	
7	3.675	17.43	
8	3.714	15.44	
9	3.753	15.42	
10	3.788	15.41	
11	3.867	14.44	
12	3.879	14.44	
13	3.956	14.39	
14	4.115	14.36	
15	4.206	13.46	
16	4.232	13.39	
17	4.252	13.36	
18	4.255	12.42	
19	4.312	11.38	knee - recommended
20	4.564	10.36	
21	4.788	10.34	
22	5.012	10.29	
23	5.690	9.29	sparsest end

Table 3. The reference approximation front: 23 non-dominated points from the high-budget run (population 48, 50 generations, seed 90909), sorted by training RMSE. No listed point dominates another; the knee is the recommended operating point discussed below.

The selected reference knee has training RMSE 4.312 mg/L, held-out RMSE 4.666 mg/L, interpretability cost 11.38, 11 active rules and 100.0% test coverage. It retains the nine mandatory coverage rules plus only two interaction rules.

Stated as a trade: moving from the knee (training RMSE 4.312 mg/L, interpretability cost 11.38) to the most accurate point on the front (3.404, cost 22.46) buys 0.91 mg/L for roughly double the interpretability cost; moving from the knee to the sparsest point (5.690, cost 9.29) saves 2.09 units of cost and gives up 1.38 mg/L. You pay X for Y is the form this recommendation has to take, because nothing in the mathematics ranks two non-dominated points against each other.

Recommendation to the shift operator

Use the starred knee as the default advisory controller. Compared with the 36-rule untuned expert system, it removes most interaction rules while improving predictive error; moving further toward the high-accuracy end of the front buys smaller error at the cost of a much larger rule set. If an operating team decides that every additional mg/L of predictive error is unacceptable, the front makes that preference explicit rather than hiding it inside one scalar weight.

Why not simply tune the scalar weight? Minimising $w_1 \times \text{RMSE} + w_2 \times \text{cost}$ slides a straight line onto the front; where the front bends, the line touches the ends and skips the middle, so some rule bases are unreachable at any weight. The scalarised GA is therefore one operating point chosen before the search, whereas the front hands the choice to the operator at the moment the preference is actually known.

Shipped rule base

R01: IF raw-water turbidity is Low AND pH is Low THEN alum dose = 41.3 mg/L.
R02: IF raw-water turbidity is Low AND pH is Medium THEN alum dose = 29.1 mg/L.
R03: IF raw-water turbidity is Low AND pH is High THEN alum dose = 39.5 mg/L.
R04: IF raw-water turbidity is Medium AND pH is Low THEN alum dose = 56.9 mg/L.
R05: IF raw-water turbidity is Medium AND pH is Medium THEN alum dose = 43.5 mg/L.
R06: IF raw-water turbidity is Medium AND pH is High THEN alum dose = 54.7 mg/L.
R07: IF raw-water turbidity is High AND pH is Low THEN alum dose = 67.5 mg/L.
R08: IF raw-water turbidity is High AND pH is Medium THEN alum dose = 57.1 mg/L.
R09: IF raw-water turbidity is High AND pH is High THEN alum dose = 65.7 mg/L.
R10: IF raw-water turbidity is Low AND 12-hour rainfall is Low THEN alum dose = 27.1 mg/L.
R15: IF raw-water turbidity is Medium AND 12-hour rainfall is High THEN alum dose = 57.0 mg/L.

Deployment boundary. This is a synthetic coursework controller. A real plant would require calibration against plant-specific jar tests, instrument uncertainty, process constraints and operating procedures before any dosing decision could rely on it.

9. Joint reflection

XU HAOWEI - “More rules must be safer”

I initially assumed that a dosing controller should keep every interaction rule because deleting knowledge sounded unsafe. Our results changed that view. The untuned 36-rule expert system averaged 6.356 ± 0.334 mg/L RMSE, while the NSGA-II knee averaged 4.804 ± 0.471 with an interpretability cost of 14.43. In the high-budget reference run, the knee used only 11 rules and still achieved full coverage. I would now separate coverage rules from optional correction rules from the beginning, instead of treating all rules as equally necessary.

XU HAOWEI - “One weighted objective is enough”

I expected a scalarised GA to be the simplest way to optimize the controller. It did produce the lower fuzzy-model RMSE (4.296 ± 0.380), but its mean interpretability cost was 17.07 compared with 14.43 for the NSGA-II knee. Seeing the front made the difference concrete: a single weight chooses the trade-off before the search, whereas NSGA-II lets the operator inspect the alternatives afterwards. Next time I would define the decision-maker and the trade-off first, then choose the optimizer.

BAO YUHANG - “The hybrid should beat a conventional model”

I assumed the more sophisticated fuzzy-evolutionary method should win on prediction. The polynomial ridge baseline disproved that: 2.155 ± 0.101 mg/L versus 4.804 ± 0.471 for the NSGA-II knee. That negative result made the project more useful because it forced us to state what the fuzzy system is for: not maximum predictive accuracy, but a compact policy whose rules can be printed, reviewed and defended. In a real study I would add a validation step that asks whether the interpretability benefit is worth the measured accuracy gap.

What we would change

- Test alternate interpretability proxies (rule count alone, rule length, overlap, semantic drift) and report sensitivity of the selected knee.
- Model process safety directly, e.g. residual treated-water turbidity or breach probability, instead of using target-dose RMSE as the accuracy objective.
- Replace synthetic distributions with plant-specific data only after checking missingness, sensor calibration and seasonal drift.
- Pre-register the operating-point selection rule so the knee is not chosen after seeing the most attractive result.

AI-use declaration

Page layout and language refinement were assisted by AI; all code execution, experimental results, figures, analysis, and final interpretation were verified by the team.